

GRAINGUARD · DESIGN DECISION RECORD

GrainGuard recommendations accepted

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DOCUMENT CONTROL

Document ID	GGD-DDR-002	Version	0.2
Status	Draft	Date	2026-09-26
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/grainguard		
Source	docs/decisions/0002-recommendations-accepted.md @ b73dd8b		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-26	Amish Chadha	Budget set to \$260 to cover the priced BOM: decided by Amish, 2026-09-26; R13 met
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all recommendations and the changes made in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below that carried a recommendation is "Decided by Amish, 2026-09-25: go with recommendation". Items without a recommendation remain "Proposed, awaiting Amish". The budget was decided by Amish on 2026-09-26 (see Budget, 2026-09-26).

■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." After GGD-DDR-001, GrainGuard had one open item with a recommendation (O3, the plenum probe), two component specifications set in GGD-CAL-001 v0.1 and flagged for Amish (a relay input of 3 mA or less and a charger clamped at 15.0 V), two open items with no recommendation (O1 and O2) and one question on the pitch wording with options but no recommendation. Where a recommendation offered several options, the recommended option is the decision. TRL 4 remains on hold by Amish's instruction, and the repo stays at TRL 3.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1 to D11	TRL 2 review items (GGD-DDR-001): budget option (a), pitch reword, control rule, RH sensing, one center cable, RS-485, solar controller with relay kit, AGM, plenum sensing if affordable, CO ₂ as an option, corn in a small Midwest bin	Already decided in GGD-DDR-001; confirmed	No change
N1 (was O3)	Adopt the plenum temperature probe despite the budget	Adopt it into the base kit and accept the \$6.50 overrun	<code>bom/bom.csv</code> item 14 is no longer an option. Per-bin kit \$247.00 to \$256.50 (2.6 % over \$250); first prototype with receiver \$269.00 to \$278.50. <code>budget_usd</code> stays \$250, and R13 records the accepted overrun (status "not met, overrun accepted"), following the recommendation to accept rather than raise the budget. GGD-CAL-001 v0.2 adds check B10: with a DS18B20 (± 0.5 °C) and the ambient SHT45 (± 0.1 °C), the fan heat is known within ± 0.51 °C and the plenum EMC within about ± 0.50 points, against a 1.85-point spread when the fan heat is only a setting. R4 moves from at risk to met on paper and is restated in GGD-REQ-001 v0.4. GGD-PRC-001 v0.4, drawing GGD-DWG-001 Rev P2 (notes), concept media (labels, key figures, flow) and <code>cad/src/model.py</code> comment updated; STEP and STL re-exported (geometry unchanged, the probe was already modeled)
N2	Relay input of 3 mA or less (flagged in GGD-CAL-001 v0.1)	Keep as specified	No change; recorded in GGD-PRC-001 v0.4, Key design choices
N3	Charger compensation clamped at 15.0 V (flagged in GGD-CAL-001 v0.1)	Keep as specified	No change; recorded in GGD-PRC-001 v0.4, Key design choices

Table 2. Items still open.

#	Item	Status
O1	First host farm and electrician for a co-design visit. No partner was recommended.	Proposed, awaiting Amish
O2	Battery and panel capacity for the recovery half of R8: a 20 W panel (about \$8 to \$10, further over budget), relax the recovery target to about 7 days, or accept the miss. No recommendation was made.	Proposed, awaiting Amish
O4	Pitch wording against the cool-mode rewet bound (at most 74 kg of water per cycle, one point in the bottom 0.37 m): keep "without rewetting it", say only "cool it or dry it", or cut the cool-mode allowance to zero. No recommendation was made.	Proposed, awaiting Amish

Table 3. Decided but on hold.

Item	Status
TRL 4 work named in the review note (bench pod and plenum probe with a lab test report, membrane equilibration time, fan heat on a real fan, relay kit fail-off timing, cold-soak of battery and charger, build log entries)	Decided as the next step, on hold: TRL 4 is on hold by Amish's instruction

No cross-repo action arises: GrainGuard does not use SwapCell or any other portfolio module.

Budget, 2026-09-26

On 2026-09-26 Amish wrote: "i approve all the budget items." Budget set to \$260 to cover the priced BOM: decided by Amish, 2026-09-26. The priced per-bin kit is \$256.50 (`bom/bom.csv` , items 1 to 11, 13 and 14, with the farmhouse receiver costed per farm under GGD-DDR-001 D1), so `budget_usd` in `project.yaml` moves from 250 to 260. This supersedes the N1 choice to keep \$250 and record the overrun: R13 moves from "not met, overrun accepted" to met. `docs/04-calcs/sizing.py` reads the budget from `project.yaml` and was rerun; GGD-CAL-001 v0.3, GGD-REQ-001 v0.5, GGD-PRC-001 v0.5, GGD-PRB-001 v0.4, `README.md` and `bom/bom-notes.md` were updated. Requirement status is now met 8, at risk 3, not met 1 (R8) and not verifiable 2.

■ Consequences

- Requirement status (GGD-CAL-001 v0.2): not met 1 (R8); not met with an accepted overrun 1 (R13); at risk 3 (R2, R6, R9); not verifiable at TRL 3 2 (R3, R10); met 7 (R1, R4, R5, R7, R11, R12, R14). Before this record: not met 1, at risk 4, not verifiable 2, met 7.
- Controlled documents bumped: GGD-PRC-001 v0.4, GGD-REQ-001 v0.4, GGD-CAL-001 v0.2, GGD-DDR-001 v0.2; drawing GGD-DWG-001 Rev P2.
- `project.yaml` : `budget_usd` stayed 250 under N1 (set to 260 on 2026-09-26, see above), pitch and problem unchanged, `trl: 3` and `trl_target: 3` unchanged.